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The Leaf Function of Penrose P2 Graphs.

All graphs considered are simple and undirected. An *induced subtree* of a graph G is an induced subgraph that is a tree. The *leaf function* of G, denoted $L_G(n)$, is the maximum number of leaves over all induced subtrees of G on n vertices; an induced subtree attaining this maximum is called *fully leafed.* A *Penrose P2-graph* is the dual (adjacency) graph of a Penrose tiling by kites and darts: its vertices are the tiles, and two tiles are adjacent whenever they share a side. *Porrier–Blondin Massé* [Int. J. Graph Comput. 1, No. 1, 1–24 (2020)] showed that $2\varphi n/(4\varphi+1) \leq L_{P2}(n) \leq \lfloor n/2 \rfloor + 1$, where φ is the golden ratio. In this paper, the authors determine the leaf function L_{P2} exactly, showing that

$$L_{P2}(n) = \begin{cases} 0 & \text{if } n \in \{0,1\}, \\ \lfloor n/2 \rfloor + 1 & \text{if } 2 \leq n \leq 18, \\ L_{P2}(n-17) + 8 & \text{if } n \geq 19. \end{cases}$$

In particular $L_{P2}(n) \sim 8n/17$. Since a Penrose tiling is aperiodic, it is perhaps surprising that L_{P2} satisfies a linear recurrence at all; the authors remark that they had expected the answer to depend on the irrationality of φ.

The upper bound builds on the result of *Porrier–Blondin Massé* [loc. cit.] that $\Delta(I) \leq 3$ for any induced subtree I of a P2-graph. The authors then prove that any 3-internal-regular induced subtree of a P2-graph has at most eight internal vertices, and is in particular a caterpillar. A minimal counterexample together with a *factorization* or *graft* operation, which splits I along an edge into two induced subtrees meeting in that edge, then yields the recurrence.

The lower bound is obtained by exhibiting a fully leafed caterpillar of every order. The base object is a caterpillar C_{116} on 116 vertices, assembled from seven copies of a caterpillar C_{14}, joined by three additional vertices at each of six junctions. That C_{116} occurs in a P2 tiling at all, and that the construction iterates, is established through the substitution structure: C_{116} arises as a φ^3-decomposition of C_{14}, and the local isomorphism property of Penrose tilings then guarantees that any patch occurring once occurs in every P2 tiling. The authors organize this by passing to the associated HBS (hexagon-boat-star) tilings, decorating their vertices according to the type of the surrounding patch, so that the caterpillars are read off as paths in this coarser tiling. The authors also remark that there exist fully leafed induced subtrees that are not caterpillars.

The authors conclude with two conjectures to the effect that the fully leafed induced subtrees are essentially sub-caterpillars of the extremal family constructed using C_{116}. The authors also note that the leaf function L_{P3} of the Penrose P3 tiling differs from L_{P2} already at $n = 12$, and ask whether fully leafed induced subtrees of the two types can be transformed into one another. Determining the leaf function for the remaining tilings of the Penrose MLD class, and for substitutive tilings generally, remains open.

The closed formula given in Proposition 3 is incorrect for $n \equiv 0, 1 \pmod{17}$ with $n \geq 36$, the smallest instances being $n = 51$ and $n = 52$, where it returns 24 in place of the correct values 25 and 26. The error occurs in the last display of the proof: writing $n = 17q + r$ with $r = n \bmod 17$, the step $L_{P2}(r + 17) = L_{P2}(r) + 8$ requires $r + 17 \geq 19$, and so fails for $r \in \{0, 1\}$, where in fact $L_{P2}(17) = 9$ and $L_{P2}(18) = 10$. The corrected values are $8q + 1$ and $8q + 2$, respectively. Shifting the residue to the range $2 \leq r \leq 18$ repairs the statement and at the same time removes the need for a separate case, giving:

$$L_{P2}(n) = \begin{cases} 0 & \text{if } n \in \{0,1\}, \\ 8\lfloor (n-2)/17 \rfloor + \lfloor ((n-2) \bmod 17)/2 \rfloor + 2 & \text{if } n \geq 2. \end{cases}$$

Some other minor typos are listed below. In Figure 4, the dark outline in the second kingdom does not correctly mark the deuce. In the algorithm of Section 4.4 the counter k is incremented at every iteration, including those in which the pointer merely advances along the derived path and no vertex is added, so that the output has fewer than n vertices; the loop should instead be controlled by the order of T. Read "The leaf function of the P2-graph" in the statement of Corollary 2. In the proof of Proposition 3, the first displayed equation should read

$$L_{P2}(n) = \lfloor (n-17)/2 \rfloor + 1 + 8 = \lfloor (n+1-18)/2 \rfloor + 9 = \lfloor (n+1)/2 \rfloor.$$

The graded poset S appearing in Conjecture 2(i) is not defined.

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